Executive Report: TikTok Data Inspection & Preparation

Project: Claim Classification Model | Client: TikTok Trust & Safety Team
PACE Phase: Analyze (Data Inspection and Structuring)

OVERVIEW

TikTok's mission is to inspire creativity and bring joy, which requires a safe and authentic environment. The platform receives a high volume of user reports regarding videos that may contain "claims" (unverified information) versus "opinions". To reduce the backlog for human moderators, we are developing a predictive model to automate the classification of this content

Business Questions

IIn this initial stage, we address specific technical and operational questions to guide the model's development:

- What is the distribution of claims vs. opinions in the current dataset?
- Are there significant differences in engagement metrics (likes, views, shares) between claims and opinions?
- What is the relationship between an author's ban status and the type of content they post?
- How does video duration impact the likelihood of a video being flagged as a claim?

Project Overview

We are transforming raw user-generated data into a structured format for machine learning. Our current workflow includes:

- Data Structuring: We successfully built a dataframe with 19,383 rows and 12 variables to ensure the information is processable.
- Dtypes Summary: We examined the data types, identifying numerical variables (int/float) for engagement and objects for categorical states (claim/verified status)..
- Anomaly Detection: We are inspecting for outliers in view counts and identifying null values that could skew the analysis.
- Feature Engineering: We are evaluating the creation of new variables, such as engagement ratios, to strengthen the model's predictive power.

PACE Methodology

We have adopted the PACE framework to ensure a disciplined data science approach:

Plan:

- Defined project goals: Building a model to streamline the claims process.
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Analyze:

- Focused on data inspection, descriptive statistics (min/max), and organizing the dataset for Exploratory Data Analysis (EDA).

Construct (Upcoming):

- Conducting full EDA and selecting features for the classification model.

Execute (Upcoming):

- Deployment of the classification tool and reporting final insights to the management team

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IMPACT

By identifying anomalies and organizing the data early, we ensure a high-quality "Tidy Data" foundation. This reduces future errors in the modeling phase and provides a clear roadmap for a Deep Dive Exploratory Data Analysis (EDA), ultimately leading to more precise claim predictions for the platform.

Key Insights

After inspecting the 12 variables in the dataset, we identified that claim_status and engagement metrics (views, likes, shares) are the most critical for the initial model. However, preliminary inspection revealed significant outliers that require cleaning before the EDA phase.

Anomalous Data Reference

- We identified "unusual values" where videos have disproportionately high view counts but zero likes or shares. This indicates the necessity of further analyzing outlier variables to prevent model bias.
- The dataset is balanced between claims and opinions (approx. 50/50), but the distribution of engagement is heavily right-skewed.
- There is a massive correlation between high engagement and account bans. Banned authors have 33x more shares (median) than active authors.
- Created normalized engagement metrics (likes_per_view, shares_per_view) to identify high-intensity content regardless of total view count.

Author Ban Status	Video View Count	Video Like Count	Video Share Count
Active	439,184	13,446	**437**
Under Review	542,399	179,305	**9,473**
Banned	743,302	224,132	**14,468**

"...suggesting that high-velocity engagement is a primary indicator for prioritized moderation review."

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Project Governance & Roles

To ensure project success, we have defined specific roles and responsibilities aligned with the PACE framework, ensuring clear communication between technical teams and institutional stakeholders.

Role	Assignee	Responsibility
Data Science Lead	Willow Jaffey	Strategic oversight, final model approval, and high-level guidance.
Data Science Manager	Rosie Mae Bradshaw	Project coordination, timeline management, and ensuring deliverable alignment.
Data Scientist	Orion Rainier	Technical mentorship, quality assurance for data processes, and advanced modeling.
Data Analyst	Marcelo Dominguez	Data cleaning, inspection, statistical analysis, and initial model construction.
Project Management Lead	Mary Joanna Rodgers	Goal validation, operational decision-making, and final tool integration.
Head of Finance / Operations	Margery Adebowale / Maika Abadi	Strategic oversight, financial goal validation, and ensuring an ethical data narrative.

Next Steps

Deep-Dive Exploratory Data Analysis **(EDA)**:

- Conduct a deep dive into the validated dataset to identify core correlations between claim_status, video_view_count, and transcription_text.

Advanced Feature Engineering:

- Prioritize variables with a direct impact on classification. We will evaluate "Data Boosters" (e.g., video sentiment and author metadata) to enrich the model's predictive power before training.

Model Selection & Evaluation:

- Evaluate Random Forest and XGBoost as primary classification candidates. We will compare performance metrics to select the most robust algorithm for automated claim detection..

Documentation & Visual Strategy:

- Update the project repository on GitHub and design specialized dashboards to communicate insights to both technical and non-technical stakeholders.

"Our analysis reveals a critical link between virality and policy enforcement. While claims and opinions are evenly split in the data, claims generate significantly higher engagement. Most importantly, the median share count for banned authors is 33 times higher than for active users. This right-skewed distribution suggests that engagement rates (per view) are the strongest predictors for identifying content that violates community guidelines."